

PP-148 PLC Control Software

Analysis & Overview Report

Project file	PP-148.SwxCF
Authoring software	Emerson PAC Machine Edition (ProficyME) 9.80 SIM 9
Company in header	Emerson Last modified by 'massner'
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How to read this report

This document explains, in plain English, what the program inside the PP-148.SwxCF project file does. The focus (as requested) is the PLC ladder logic, the hardware / I/O arrangement, and the way the operator's touchscreen (HMI) talks to the PLC. A separate appendix at the end lists every tag database group and the major instructions the ladder uses, so you can refer back when you open the project.

A few words up front. PLC is short for Programmable Logic Controller — a rugged industrial computer that reads switches and sensors, runs a cyclic program, and turns outputs on and off. The program in your file runs on a GE Series 90-30 PLC (one of the most widely deployed industrial controllers of the 1990s and 2000s). The operator interface is a 12-inch QuickPanel View touchscreen. Everything the operator sees and most of what moves on the machine is coordinated by the PLC program you uploaded.

1. What the file actually is

PP-148.SwxCF is an Emerson PAC Machine Edition project archive. Under the hood it is a Microsoft Compound Storage File (the same container format as a classic .doc file), and inside that container live more than two hundred named streams. Each stream is a sub-document: the PLC ladder block, the tag database, the HMI panels, the alarm groups, the communication driver configuration, and so on.

The companion PP-148.xml file is simply the project's 'manifest' — it tells PAC Machine Edition which version authored the file, and what targets the project contains. There are two targets:

- PP3V80 — a GE Series 90-30 PLC. This is the brain of the machine and runs the ladder program.
- Pipepro_Screen — a 12-inch QuickPanel View TFT (model IC754Vxx12Cxx) running Windows CE. This is the operator touchscreen and is named 'PP3-viewscreen' inside the project.

The header text inside the project also mentions 'MICHIGAN' and an e-mail address at hawkeye-pipe.com — Hawkeye Pedershaab, concrete pipe machine builder in Mediapolis, Iowa (now part of Afinitas). Reading the schematics and 2022 condition report you supplied together with the program, PP3 is a Hawkeye Pipe Pro III 'Standard Image' three-station packerhead concrete pipe machine, and the software in PP-148 is the controls package for that machine.

2. The big picture — what the program controls

A Pipe Pro III builds reinforced concrete pipe vertically, one at a time, on a turning carousel with three pipe-forming stations. Wet concrete is dropped from an overhead chute through a short belt conveyor into the form; a rotating 'packerhead' plunges down into the form and compacts the concrete against the jacket while spinning. When the pipe is full, the carousel indexes the newly-made pipe to an unload position and presents the next empty form to the packerhead. A transfer cart then carries finished pipe headers out to a handling table.

In terms of what the PLC physically makes happen, the program coordinates:

- Concrete feed — hopper agitator, feed belt forward/reverse, concrete-on-belt detection, and a rotating feed chute.
- Packerhead cycle — head up/down, clamp/unclamp, full-pressure, trowel forward/reverse, top and bottom form vibrators, and a fill-check routine.
- Carousel — a servo-positioned three-station turntable (Stations A, B, C, plus Home) with lock pins that drop in at each station, a high-speed encoder for feedback and an Axis Positioning Module (APM) driving the servo.
- Turntable (T-TABLE) — a second rotational axis with its own high-speed counter and homing routine.
- Header transfer cart — a shuttle between the machine and an ergonomic handling table (the 'ERG TABLE') with lock pins and multi-speed motion.
- Retain rings and head stop pins — hydraulic support rings that hold the steel reinforcing cage, plus pins that index and secure heads.
- Oil-dip cycle — an automatic release-agent dip cycle for the packerheads at form-change time.
- Alarm & horn system — horn on/off timers, three alarm registers, acknowledge/silence and test.
- Recipe storage — the operator saves and recalls 'form data' per carousel station (a pipe recipe: timers, fill factor, speeds, etc.).
- Supervision — real-time comms to a sub-controller labelled 'PF' (Pallet Feeder), an automatic logon, time-of-day clock, and diagnostic bits.

3. Hardware & I/O

3.1 The PLC platform

The target is a GE Fanuc / Emerson Series 90-30 modular PLC. On a 90-30 the controller is a CPU module plugged into a backplane rack along with a mix of power supply, discrete I/O, analog I/O, and 'intelligent' option modules (serial, motion, high-speed counter, etc.). The CPU scans a ladder program in a loop: read inputs, solve ladder, write outputs, service communications — typically every few milliseconds on the 90-30 series.

The project's hardware configuration is stored in a stream called Hardware.hwcfg inside the archive; that stream is compressed and proprietary, so the rack layout is not readable from text alone. What can be read directly from the tag database is the full I/O address map the program uses.

3.2 The tag database (what the PLC watches and controls)

The project defines roughly 424 named tags, organised by the Series 90-30 reference table they live in. The following summary lets you see the shape of the program at a glance:

GE memory	Count	What it is	Examples
%I (discrete inputs)	39	Physical switches, limit switches, proximity sensors, pushbuttons, key switches, relay contacts coming from the field.	I10–I120
%Q (discrete outputs)	32	Solenoid valves, motor starters, indicator lamps and the horn — things the PLC switches on and off.	Q10–Q100
%AI (analog inputs)	4	Continuous values from sensors such as pressure transducers, position (linear) transducers, or analog feedback. Two of the four are declared DINT (32-bit), typical of high-resolution position sensors.	AI2, AI3, AI4 [DINT], AI10 [DINT]
%AQ (analog outputs)	4	Continuous set-points sent out, e.g. a drive reference or a proportional-valve command.	AQ1–AQ4
%M (internal bits)	136	Internal coils — the PLC's own scratch-pad of true/false values, one bit per rung step, sequencer, or latched state.	M129–M977
%R (registers)	184	16-bit working registers for timers, counters, set-points, math results. Several are declared DINT (32-bit) and are used for position, totalised counts, and time accumulators.	R25, R75, R90, R105, R209, R250...
%SA (system bit)	1	One 'SA' (system / diagnostic) reference — these mirror CPU fault states.	SA11

Two other register groups are defined implicitly by the program. There are about a dozen %G (global) references inside the COMM_REQ parameter blocks (used when the CPU talks to an intelligent module), and the 90-30's %T (temporary) table is not used in this project.

3.3 Intelligent / special modules

Three 'COMM_REQ' (communication-request) function blocks fire from the program at rungs 15/18, 51/54 and 148/150. Each one is preceded by a BLKMOV_WORD that stuffs a parameter block into an R-register and then a COMM_REQ hands that block to a module in a specific rack / slot. This is the standard pattern on the 90-30 for configuring 'smart' modules at startup. Based on the tag comments, the three COMM_REQs correspond to:

- An APM (Axis Positioning Module) that drives the carousel servo — tags like 'CAROSEL APM COMMAND DATA', 'CAROSEL FRM APM ACTUAL POSITIN', 'CAROSEL APM DATA CODE', and 'CAROSEL TO APM CLEAR ERROR' confirm this.
- A High Speed Counter (HSC) module that reads encoder feedback for both the carousel and the T-TABLE — 'FRM HSC CAROSEL POSITIN / SPEED', 'FRM HSC T-TABLE POSITIN / SPEED', 'HSC MODULE READY', 'HSC COM REQUEST FAULT'.
- A CMM (serial communications) module whose 'PF' port talks to the Pallet Feeder sub-controller — 'FM PF', 'FRM PF', 'PALLET FEEDER COMM TIMER', 'NO PF COMM. FAULT TRIGGER'.

A single SVC_REQ instruction at rung 841 uses service-request function 7. On the 90-30, function 7 is 'Read / Change the PLC time-of-day clock'. The program uses it to keep the real-time clock in sync (you

will see 'DAY OF WEEK BCD', 'MONTH BCD', 'HOURS BCD', 'MINUTE / SECOND BCD4' registers and a 'DAY OF WEEK CHANGED TRIGGER' — all of them get loaded from this service request).

4. The ladder program

4.1 Size and shape

The entire logic lives in a single Ladder Diagram block named `_MAIN` with 860 rungs and approximately 4,100 instructions. There are no subroutines, function-block calls, or structured-text blocks — it is deliberately a flat, top-to-bottom ladder that scans in sequence on every CPU pass. That is typical of pipe-machine control from this era: easy for maintenance technicians to probe with a handheld and easy to troubleshoot with a printout.

Instruction mix (from the cross-reference table embedded in the file):

Instruction family	Count	What it does in this program
Relay / bit logic (contacts, coils, SET, RESET, transitions)	3,072	By a long way the dominant work of the program: reading switches and internal bits, latching state, setting and resetting flags. 1,762 normally-open contacts + 548 normally-closed + 377 plain coils + 140 SET coils + 171 RESET coils + 70 positive-edge coils tell you this is a state-machine-heavy sequence controller.
Integer and double-integer math / compare	843	Add, subtract, multiply, divide, modulo, equal, not-equal, greater/less-than on 16-bit and 32-bit integers. Used throughout for time accumulators, speed calculation, position comparisons against station limits, fill-factor math, and recipe arithmetic.
Timers	121	Tenths-of-a-second (<code>TMR_TENTHS</code> , 105 uses), hundredths-of-a-second (<code>TMR_HUNDS</code> , 16 uses) and on-delay timers (<code>ONDTR_TENTHS</code> , 11 uses) drive the cycle timing — fill time, vibrator time, dwell times, alarm-horn on/off, retract delays, etc.
Counters	7	All up-counters (<code>UPCTR</code>). They total pipes produced, index counts, key-set changes, and similar.
MOVE / Block move / Array move	309	<code>MOVE_INT</code> (275), <code>BLKMOV_WORD</code> (6) and <code>ARRAY_MOVE_INT</code> (28) shuffle set-points and recipes between R-registers — used heavily by the form-recipe save/load routine.
Word bit operations (<code>AND_WORD</code> , <code>SHIFTR_WORD</code>)	24	Extract bit fields out of the feeder's key-selector, build BCD display values for the HMI, etc.
BCD conversions (<code>BCD4_TO_INT</code>)	24	Convert Binary Coded Decimal values (typical of thumbwheel-style or clock registers) to normal

Instruction family	Count	What it does in this program
		integers so they can be compared or displayed.
COMM_REQ (intelligent-module requests)	3	Configure APM, HSC and CMM (serial) modules at startup — see §3.3.
SVC_REQ 7 (time-of-day clock)	1	Keeps the real-time clock current.

A single rung comment survives in the binary: “THE FOLLOWING RUNGS SEND DATA INFO TO THE PALLET FEEDER.” That matches the COMM_REQ immediately below it — a block of pipe-recipe data being pushed down the serial link to the Pallet Feeder controller at the start of each cycle.

4.2 How the rungs are organised (inferred)

Reading the cross-reference and the tag-name groupings together, the 860 rungs naturally cluster into the following 'phases' — a common ladder pattern for a cyclic machine. Exact rung ranges are from the cross-reference; they are approximate because there are no ladder-folders defined at this level, only a single block.

Rungs	Phase	What happens
1 – 50	Power-on / initialisation	System bits, E-STOP scan, home-status, first-scan BLKMOV_WORD + COMM_REQ to configure the APM.
51 – 150	Module start-up & PF handshake	Second and third COMM_REQ (HSC and CMM/Pallet-Feeder), watchdog timers on each, and first communication to 'PF'.
150 – 260	Operator key switches & permissives	Auto/Manual key switch, supervisor keys (FEEDER / OIL-DIP / PICK-UP / P-HEAD), E-STOP chain, and 'DISABLE AUTO FUNCTIONS' from the HMI.
260 – 440	Carousel / T-TABLE motion	APM speed & deceleration set-ups, 'CAROSEL ENABLE FWD SLO SPEED', 'CAROSEL LCK PIN EXTEND / RETRACT', station position compares, 'CAROSEL HOME DETECTD', 'CAROSEL IS MOVING', 'PHEAD CAROUSEL LOST ITS POSITION' alarm, plus the symmetric T-TABLE routine.
440 – 560	Packerhead (P-HEAD) up/down/clamp	Per-station clamped/unclamped status, full-pressure down, heads-up, float, trowel forward/reverse, top and bottom vibrators during fill, 'CHECK FILL', 'FORM PRE-FUL' and 'FORM FULL & DONE' logic.
560 – 690	Feed belt / chute / concrete detection	'BELT FORWARD SWITCH', speed multipliers, 'CONCRET ON FEED BELT' timers, belt-off trigger, chute rotation, hopper agitator forward/reverse.
690 – 770	Header cart & pick-up	'CART FWD', 'CART IS AT MACHINE / AT ERG TABLE', 'CART PINS 1-5', pick-up clamp / float / retain-rings cycle, and the 'PICK-UP SEQUENCE DONE' flag.
770 – 830	Oil-dip and recipe	Oil-dip supervisor-key permissives, heads-up/down/full-press/unclamp routines; save/load/delete form recipe by station (STA.A / STA.B / STA.C).

Rungs	Phase	What happens
830 – 860	Alarm horn and housekeeping	Alarm horn ON/OFF cycle, alarm-test, SVC_REQ 7 clock update, HMI 'FM SS' (from screen) clear-alarms / test-alarms handshake, final housekeeping bits.

4.3 Notable design patterns

- **Hardware counters, not software loops** — The carousel and turntable positions are read from an HSC module, not counted in ladder. The ladder only asks the HSC for the current value ('FRM HSC CAROSEL POSITIN') and compares it against stored per-station limits. This is the only way to reliably track a servo on a 90-30.
- **Motion-profile arithmetic in ladder** — Set-points for 'CAROSEL CURRENT FWD SLO POSITIN', 'PRE STP VALUE' and 'DECEL STEP VALUE' are computed from station position and operator-entered speeds, then pushed to the APM via MOVE_INT. In effect the ladder acts as a very simple motion planner.
- **Recipe storage per station** — 'FORM DATA STORAGE START', 'FORM DATA STORAGE LOC #2', 'CURRENT FORM ID TO LOAD / SAVE / DELETE', 'LOAD MATCH FOUND', 'NO SPACE TO SAVE DATA' and the ARRAY_MOVE_INT instructions are a small in-memory recipe database. Each entry stores the timers, fill factor, and speeds for one pipe size; the operator selects a station and loads or saves a recipe on it.
- **Horn and lamp cycling** — 'ALARM HORN CYCLE BIT', 'NORMLIZ LIGHT CYCLE BIT', 'INDX PB LIGHT TOGGLE BIT' and paired ON/OFF timers implement simple blink patterns on lights and horns without any hardware flasher.
- **Pallet Feeder as a separate controller** — Tags starting 'FM PF' and 'FRM PF' (E-STOP OFF, JIB & BOOM HOMED, OK TO ROTATE MACHINE, PALLETS NOT LOADING, ALARM SILENCE, RUN L/U VIBS) show the main PLC is NOT the only controller on the machine — it interlocks with a pallet-feeder PLC over the serial link, and neither machine will run until both sides confirm safe status.

5. Communication — HMI to PLC

The operator touchscreen talks to the PLC over a point-to-point SNP (Series Ninety Protocol) serial link. SNP is the native protocol of the GE 90-30 family. The driver configuration stored inside the project spells out the wire-level settings exactly:

Parameter	Value
Driver	SNP (Driver12) on the QuickPanel View
Physical interface	RS-232
Port	COM2 on the QuickPanel
Baud rate	19,200
Data bits / parity / stop	8 data, odd parity, 1 stop
Handshaking	None
Topology	Point-to-Point (single PLC on the cable)
Device name	PP3 (matches the PLC target name PP3V80)

Parameter	Value
Scan rate	1,000 ms (driver polls tags once per second on average)
Max bytes / transaction	500
Transaction timeout	2,000 ms

Settings of 19,200 / 8 / Odd / 1 on COM2 match exactly the factory defaults used on Series 90-30 port 1 when the CPU is set for SNP master/slave. The driver definition also includes a 'Driver6' entry with no devices attached — this is the HMI's internal 'system' driver for tags like #AccessLevel, #CurrentTime, #DriverReads, etc.

5.1 The HMI (QuickPanel View) project

The operator interface is a panel called Pipepro_Screen. It contains about nineteen screens (referred to internally by generic names such as 'UniqueName11') plus a persistent Header bar and a Home panel. Short VBScript-style fragments attached to each screen call a built-in function named StringCopy to write the current screen's name into a shared tag, e.g. 'STATUS1' or 'MANUAL2'. That makes the PLC aware of which screen the operator is looking at — useful for enabling the correct manual-jog buttons, for example.

Screen group	Members	Purpose
Header	shared top bar	Displays the active screen name, date/time, alarm count, active-language flag.
Home / Menu	Home, Alarm1	Main navigation and alarm summary window.
Status screens	STATUS1 – STATUS5, STATUS_OVERVIEW	Live read-outs: pipe count, carousel position, current cycle timer, sub-system state. A shared script pushes the active label into 'status_save'.
Manual screens	MANUAL1, MANUAL2, MANUAL3	Jogging and direct-output control — the dense cluster of 'Ellipse' touch-animations (over seventy on MANUAL1 alone) gives each valve / motor a dedicated button. 'man_save' tracks which one is active.
Machine setup	MS1, MS2, MS3, MS4	Set-point and numeric data entry for speeds, pressures, vibration times — 'ms_save' records which MS screen is active.
Pipe / form setup	SETUP1, SETUP3	Recipe screens — save / load the form ID for each carousel station; 'setup_save' tracks the active setup screen.

An application-level logon script dated 5 September 2002 auto-logs the user as 'Oper' at startup, so the operator does not have to type credentials to see the Status / Home screens (supervisor functions still require a key-switch, see §6).

6. Alarms

Alarms are defined in three groups inside the HMI project — Bit Alarms (digital), Word Alarms (numeric threshold) and a Root group. The severity levels used are the standard PipePro set: LOLO, NRML, HIHI, MIN%, MAJ%, MINV, MAJV, EVENT, ALRM, plus an Event-type pseudo-level.

The ladder side of alarming is built on three patterns you will see repeatedly:

- A 16-bit 'ALARM STORAGE REGISTR' plus three 'ALARM REG #n HAS CHANGED' triggers — each bit in the register represents one alarm and the ladder detects change-of-state to drive the horn.
- A pair of timers ('ALARM HORN ON TIMER' / 'ALARM HORN OFF TIMER') and the 'ALARM HORN CYCLE BIT' generate a pulsed horn output until acknowledged.
- Dedicated 'FAIL TO MOVE' timers attached to each axis or cylinder — 'CAROSEL LCK PIN ALARM TIMER', 'CHUTE NOT ROT ALARM TIMER', 'HEAD STOP PIN DID NOT EXT/RET', 'HYD PUMP ALARM TIMER', 'LID NOT DOWN ALARM TIMER', 'NO FORM DETECTED ALARM TIMER'. If the expected motion doesn't complete inside its timer window, the alarm latches.

The HMI side acknowledges by writing to specific bits the PLC watches — 'FROM SS CLEAR ALARMS', 'FROM SS ALARM SCREEN KEYS', 'FROM SS TEST ALARMS', 'FM PF ALARM SILENCE'. This two-way handshake (PLC latches the alarm, HMI clears it) is the correct pattern and prevents a screen glitch from silently clearing a real fault.

7. Observations and practical notes

- **Relatively recent 2026 timestamp on a 1990s machine** — The project file was last modified in February 2026 (by user 'massner' at Emerson), even though the machine is a 1994-era Pipe Pro III. The 2022 Afinitas service report you supplied specifically recommends “a full controls upgrade would be required”. That is almost certainly what PP-148 is — the modern retrofit of the original controls, brought forward into PAC Machine Edition 9.80 but still targeting the Series 90-30 CPU and QuickPanel View so the customer can retain familiar hardware.
- **Ageing hardware to keep an eye on** — The Series 90-30 CPU, the SNP serial driver and the QuickPanel View (Windows CE) are all in extended support / end-of-life from Emerson. If this is a pre-upgrade project file, the migration path Emerson typically recommends from here is to an RX3i CPU in the same rack (keeps the I/O) and a newer QuickPanel+ panel running the same PAC Machine Edition project.
- **One single ladder block is both a strength and a risk** — Having all 860 rungs in _MAIN is easy to print, search and troubleshoot, but it also means every rung is evaluated on every scan. If the program grows further the CPU will eventually run out of scan time. A future refactor into subroutines by subsystem (CAROSEL, P-HEAD, FEED BELT, RECIPE, ALARMS) would make maintenance far easier without changing behaviour.
- **Comments are almost entirely in variable descriptions, not rungs** — Only one rung comment survived into the binary ('THE FOLLOWING RUNGS SEND DATA INFO TO THE PALLET FEEDER'). That is normal for GE projects because rung comments are stored in a separate stream and PAC Machine Edition only keeps them if they are explicitly enabled. The descriptive meaning of the program lives in the tag names themselves — which is why the tag database is so unusually rich.
- **Safety chain is distributed** — The emergency-stop chain runs through the Pallet Feeder PLC first ('E-STOP OFF RELAY FROM PF' and 'FRM PF E-STOP OFF') and then into the main PLC. Anyone working on either controller needs to know this — dropping the E-STOP on the PF machine will stop PP3 too, and vice-versa.

- **Hardware-configuration stream is compressed** — The exact rack/slot layout (which slot is the APM, which is the HSC, which holds the DC-in 32-point card, etc.) is inside 'Hardware.hwcfg' and is not plain text. To read it reliably you must open the project in PAC Machine Edition itself — double-click Hardware Configuration under the PP3V80 target.

Appendix A — GE 90-30 instruction glossary (only what this program uses)

Instruction	What it does
NOCON / NCCON	Normally-Open / Normally-Closed contact. Reads a bit; conducts power when true/false.
COIL	Assigns the power-flow state at the end of the rung to a bit (non-retentive).
SETCOIL / RESETCOIL	Latch a bit on / off; retained until the opposite operation.
POSCOIL / NEGCOIL	Positive / negative edge coil — fires for exactly one scan when the rung transitions.
NCCOIL	Negated coil — inverse of COIL.
TMR_TENTHS / TMR_HUNDS	Timer counting in 0.1-second / 0.01-second increments.
ONDTR_TENTHS	On-delay timer (must stay true for $N * 0.1$ s before coil true).
UPCTR	Up-counter — increments once per positive edge on its enable input.
MOVE_INT	Copy a 16-bit value.
BLKMOV_WORD	Copy a fixed block of words to a destination (used to stage COMM_REQ parameters).
ARRAY_MOVE_INT	Copy N consecutive 16-bit words from one array to another (used by the recipe store).
ADD / SUB / MUL / DIV / MOD (_INT / _DINT)	16-bit or 32-bit integer math.
EQ / NE / GT / LT / GE / LE (_INT / _DINT)	Comparison instructions; pass power if the relation is true.
AND_WORD / SHIFTR_WORD	Bit-wise AND and right-shift on 16-bit words — used mostly to extract BCD / key-switch bit fields.
BCD4_TO_INT	Convert a 4-digit BCD word to a standard integer.
COMM_REQ	Communication request — send a pre-built parameter block to a smart module (APM, HSC, CMM).
SVC_REQ	Service request — call a CPU firmware service. Function 7 in this program = read / set time-of-day clock.

Appendix B — Subsystem tag groups

The tag database is rich — names tell you what the program is doing. Below is a grouped roll-up; the full tag dictionary is inside the project file itself. Roughly 80 tags begin with CAROSEL, 30 with P-HEAD, 25 with CART, 20 with T-TABLE, 18 with FEEDER, 14 with PICK-UP, 12 with HSC and 11 with PF.

Family	Representative tags (abbreviated)
Carousel motion	CAROSEL ENABLE FWD SLO SPEED, CAROSEL IN POS CHECK VALUE, CAROSEL LCK PIN EXTEND / RETRACT, CAROSEL HOME DETECTD, CAROSEL LOCKED @ STA.A/B/C, CAROSEL LOST POSITIN, CAROSEL APM COMMAND DATA, CAROSEL TO APM CLEAR ERROR
Turntable motion	T-TABLE FWD HI SPEED, T-TABLE HOMING FWD/REV SPD, T-TABLE STOP SPEED, FRM HSC T-TABLE POSITIN / SPEED
Packerhead cycle	P-HEAD CYCLE ENABLED, P-HEAD CYCLE DONE, P-HEAD FULL PRESS DOWN, P-HEAD ENABLE TOP/BTM VIBS, P-HEAD RETAIN RING UP/DOWN ENABLE, P-HEAD STA.A/B/C CLAMPED / UNCLMPD / AT DOWN / AT UP / TROWEL BACK
Pick-up / retain rings	PICK-UP HEADS UP PB, PICK-UP HEADS CLAMP PB, PICK-UP DELAY 2 CLAMP TIMER, PICK-UP HDS DWN AND CLAMPED, PICK-UP UNCLAMP OVERRIDE
Header transfer cart	CART FWD HI / LOW SPEED, CART IS AT MACHINE / AT ERG TABLE, CART PINS 1–5, CART STOPPED MOVING TIMER, HEADER CART ACCEL / DECEL TIME
Concrete feed	BELT FORWARD / REVERSE SWITCH, BELT SPEED MULT. FACTOR, CONCRET DETECTR, CONCRET ON FEED BELT, CHUTE FORWARD / REVERSE, CHUTE HOME TRIGGER, HOPPER AGITATR FORWARD / REVERSE
Feeder (pallet feeder)	AUTO FEEDER EXTEND / RETRACT ON, FEEDER BTM / TOP VIB AUTO, FEEDER TRAY FORWARD / EXTEND / BACK, FEEDER LID DID NOT CLOSE, FEEDER KEY SET #0..#3, FEEDER SUPVSR KEY ACTUATD, PALLET FEEDER COMM TIMER
Oil-dip routine	OIL DIP ENABLE HEADS FLOAT / RAISE HEADS, OIL DIP HEADS ARE UP / DOWN, OIL DIP DONE, OIL-DIP SUPRVS KEY ACTUATD
Recipe store	CURRENT FORM ID @ FEEDR, CURRENT FORM ID TO LOAD / SAVE / DELETE, DATA SAVED / DELETED, LOAD / DELETE FORM ID FIND DONE, NO SPACE TO SAVE DATA
Alarms / horn	ALARM HORN ON / OFF TIME / TIMER / CYCLE BIT, ALARM STORAGE REGISTR, ALARM REG #n HAS CHANGED, HYD PUMP ALARM TIMER, LID NOT DOWN ALARM TIMER, NO CONCRET TIMER, HSC COM REQUEST FAULT
Real-time clock	HOURS BCD, MINUTES BCD, MINUTE / SECOND BCD4, DAY OF WEEK BCD, DAY OF MONTH / HOUR BCD4, MONTH BCD, MONTH / DAY CHANGED TRIGGER
HMI (SS / Smart Screen)	FROM SS DISABLE AUTO FUNCTIONS, FROM SS CLEAR / TEST ALARMS, FROM SS CART FWD / REVERSE, FROM SS SAVE / DELETE STA.A/B/C, FROM SS ROTATE HEADS REVERSE, FROM SS START / STOP OIL DIP
Pallet-Feeder link (PF)	FM PF ALARM SILENCE, FRM PF COMM BIT, FRM PF E-STOP OFF, FRM PF JIB & BOOM HOMED, FRM PF OK TO ROTATE MACHINE, FRM PF PALLETS NOT

Family	Representative tags (abbreviated)
	LOADING, FRM PF RUN L/U VIBS

Appendix C — Schematics and the 2022 condition report

The five Hawkeye drawings you uploaded confirm the machine identity and layout:

Drawing	Revision / date	Subject
22432D	Issue C, 1988 (CADised 1994, Rev C 1998)	Pipe-Machine Hydraulic Unit Placement Clearances — the hydraulic power unit (tank, motor, return-line filter, thermometer) feeding the pipe machine.
25769C	Issue A, 16-May-1991	PPIII — Handling Instructions for Pipe-Pro Machine Components — rigging / lifting guide for the modules.
29551D	Issue A, 03-Jun-1994	Pipe Pro III Standard-Image Machine, 0° Feeder, with In-line Header Transfer Cart — the overall installation layout of the variant of the machine with a straight-up feeder.
29878D	Issue C, 29-Aug-1994 (mod 2010)	Pipe Pro III-18° Standard-Image Machine Installation Illustration — the variant with an 18° tilted feeder.
29955C	Issue A, 27-Sep-1994	Pipe Pro III Machine Component Placement for Peace Dale Precast — site-specific layout for one customer.

The 'Hawkeye 29551' drawing in particular shows the footprint that matches the tag database one-for-one: three-station carousel, in-line header transfer cart to an ergonomic unloading table, overhead rotating chute above a conveyor belt, and a hydraulic power unit feeding the machine — exactly the devices the PLC's CAROSEL, CART, CHUTE, BELT and HYD PUMP tag groups target.

The 2022 Afinitas service report (Stoller / Holtkamp, 2-4 August 2022) inspected the Jensen / Van Cleve Pipe Pro III forms ranging from 12" Quad up to 48" Single. That report's conclusion — 'I believe a full controls upgrade would be required' — dovetails with the 2026-dated project file you supplied, strongly suggesting PP-148 is the replacement controls bundled with the refurbishment.

Closing

If you want any part of this expanded into a deep-dive — for example a rung-by-rung walk-through of a particular phase, an I/O-point worksheet mapped to the physical devices shown on the schematics, or a migration plan off the Series 90-30 — just say which and I will produce it.